PAIR CORRELATION OF ZEROS OF THE RIEMANN ZETA FUNCTION I: PROPORTIONS OF SIMPLE ZEROS AND CRITICAL ZEROS

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Dedicated to Hugh Montgomery, whose insight continues to enlighten and surprise.

Abstract. Assuming the Riemann Hypothesis (RH), Montgomery proved a theorem in 1973 concerning the pair correlation of zeros of the Riemann zeta-function and applied this to prove that at least 2/3 of the zeros are simple. In this paper, we investigate the versatility of the pair correlation method and show, for the first time, that it can be used to prove results on the *horizontal distribution* of zeros of the Riemann zeta-function.

In earlier work we showed how to remove RH from Montgomery's theorem and, in turn, obtain results on simple zeros assuming conditions on the zeros that are weaker than RH. Here we assume a more general condition, namely that all the zeros $\rho = \beta + i\gamma$ with $T < \gamma \le 2T$ are in a narrow vertical box centered on the critical line with width $b/\log T$, where $b \to 0$ as $T \to \infty$. We first prove the generalization of Montgomery's result that at least 2/3 of zeros are simple, and we then prove the new result that the pair correlation method yields at least 2/3 of the zeros on the critical line. We also use the pair correlation method to prove that at least 1/3 of the zeros are both simple and on the critical line, a result already known unconditionally using different methods.

1. Introduction and Statement of the Main Results

Let $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. A central problem in analytic number theory is to quantify the proportion of zeros of the Riemann zeta-function, $\zeta(s)$, that are simple and on the critical line $\Re(s) = 1/2$.

In his now famous paper on the pair correlation of the zeros of the Riemann zeta-function, Montgomery [Mon73] introduced a method, conditional on RH, designed to study the vertical distribution of the zeros of $\zeta(s)$. Under RH, Montgomery [Mon73] showed that more than 2/3 of the zeros are simple, and soon after, Montgomery and Taylor [Mon75] improved the proportion to 67.25%. In 2020, Chirre, Gonçalves, and de Laat [CGdL20] used Montgomery's method to show that, under RH, more than 67.92% of the zeros are simple.

In 1974, Levinson [Lev74] introduced an unconditional method designed to study critical zeros, and he proved that more than 1/3 of the nontrivial zeros are on the critical line, which he improved soon thereafter [Lev75] to more than 0.3474. Heath-Brown [HB79] and Selberg (unpublished) subsequently showed that more 34.74% of the nontrivial zeros are on the critical line *and* simple. As in the case of Montgomery's method, Levinson's method is still actively used. Indeed, in 2020 Pratt, Robles, Zaharescu, and Zeindler [PRZZ20] proved that more than 41.72% of the zeros are on the critical line and that more than 40.75% of the zeros are on the critical line and simple. For a detailed discussion of the rich history of Levinson's method and its modifications, we refer the reader to the introduction of [PRZZ20].

In 1998, Conrey, Ghosh, and Gonek introduced a method which uses the discrete, mollified moments of $\zeta'(\rho)$ to study simple zeros. They showed on RH and an additional hypothesis that at

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least $19/27 = 70.3\overline{703}\%$ of the zeros are simple. Later Bui and Heath-Brown [BHB13] showed that this result holds on RH alone.

1.1. **Montgomery's method and critical zeros.** In the recent paper [BGSTB24], we returned to Montgomery's pair correlation method and under an assumption weaker than the Riemann Hypothesis, showed that at least 61.7% of the zeros are simple. There, we wrote "The method of proof neither requires nor provides any information on whether any of these zeros are or are not on the critical line where $\beta = 1/2$." In this paper, however, we show that this statement is incorrect. In particular, assuming that the zeros of $\zeta(s)$ are contained in a narrow vertical box centered on the critical line, we prove the new result that the pair correlation method yields at least 67.25% of the zeros on the critical line and that at least 34.5% of the zeros are both simple and on the critical line. We give a sketch of the ideas of the proof below and then precisely state our results in Theorem 1 and Theorem 2.

Remark 1. Comparing numerical results, we see that the pair correlation method obtains a stronger proportion of critical zeros than does Levinson's method, with the obvious caveat that the pair correlation method requires an assumption on the zeros of $\zeta(s)$ and Levinson's method is unconditional. On the other hand, the arguments presented here reveal a novel observation: the pair correlation method, which was developed to study *vertical distribution* of zeros of $\zeta(s)$ can be used to prove results on the *horizontal distribution* of zeros of $\zeta(s)$ as well.

1.2. **Sketch of the approach.** Assuming RH, Montgomery applied the pair correlation method to prove that there are simple zeros by using the multiplicity weighted diagonal terms from the correlation of each zero ρ with itself. This sum is bounded by a kernel weighted sum over the differences of all the pairs of zeros. With a Fejér kernel this bound is 4/3 of the number of zeros, and from this bound one easily obtains that at least 2/3 of the zeros are simple.

As we make explicit below, when RH is not assumed one also obtains *symmetric* diagonal terms from the correlations of each zero ρ with its symmetric pair $1 - \overline{\rho}$, which only occur for zeros off the critical line with $\beta \neq 1/2$. The same 4/3 of the number of zeros now bounds the sum of the diagonal terms plus the symmetric diagonal terms when we assume all the zeros are in a thin vertical box. Since the diagonal terms trivially have the number of zeros as a lower bound, the number of symmetric diagonal terms can be at most 1/3 of the number of zeros. Thus, as we show in Theorem 1, at least $2/3 = 66.\overline{6}$ of the zeros must be on the critical line. Using a more complicated kernel due to Tsang [Tsa93], we can improve these numerical results and show that at least 67.25% of the zeros are on the critical line.

1.3. **Statements of results.** For $b > 0$, let B_b be the thin vertical box centered on the critical line defined by

$$B_b := \left\{ s = \sigma + it \;\middle|\; \frac{1}{2} - \frac{b}{2\log T} < \sigma < \frac{1}{2} + \frac{b}{2\log T},\ T < t \le 2T \right\}, \tag{1.1}$$

and let $N(B_b)$ denote the number of zeros of $\zeta(s)$ in B_b. Let $N_s(B_b)$ count the number of zeros in B_b that are simple, and let $N_0(B_b)$ count the number of zeros in B_b that are on the critical line. Finally, let $N_0^s(B_b)$ denote the number of zeros in B_b that are simple and on the critical line. Note that in each of these settings, we count zeros with multiplicity.

Theorem 1. *Assume that, for all sufficiently large T, all the zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $T < \gamma \le 2T$ are in B_b. Then we have, where $b \to 0$ as $T \to \infty$,*

$$N_s(B_b) = \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \rho \text{ simple}}} 1 \ge \left(\frac{2}{3} + o(1)\right) \sum_{\substack{\rho \\ T<\gamma\le 2T}} 1 = \left(\frac{2}{3} + o(1)\right) N(B_b), \tag{1.2}$$

$$N_0(B_b) = \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta=\frac{1}{2}}} 1 \ge \left(\frac{2}{3}+o(1)\right) \sum_{\substack{\rho \\ T<\gamma\le 2T}} 1 = \left(\frac{2}{3}+o(1)\right) N(B_b), \tag{1.3}$$

and

$$N_0^s(B_b) = \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta=\frac{1}{2} \text{ and } \rho \text{ simple}}} 1 \ge \left(\frac{1}{3}+o(1)\right) \sum_{\substack{\rho \\ T<\gamma\le 2T}} 1 = \left(\frac{1}{3}+o(1)\right) N(B_b). \tag{1.4}$$

Remark 2. The result in (1.4) is known unconditionally and was first proved independently by Heath-Brown [HB79] and Selberg (unpublished), see [Tit86, Sections 10.28 and 10.29].

We state the results in the following theorem for two choices of fixed b, not simply as a statement of proportion, but rather utilizing the asymptotic formula for the number of zeros ρ with $T < \gamma \le 2T$, which is given in (2.6) below.

Theorem 2. *Assume that, for all sufficiently large T, all the zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $T < \gamma \le 2T$ are in B_b. Then as $T \to \infty$, we have*

$$\begin{aligned} N_s(B_{0.3185}) &\ge \left(0.66666908 + o(1)\right) \frac{T}{2\pi} \log T, \\ N_0(B_{0.3185}) &\ge \left(0.66666908 + o(1)\right) \frac{T}{2\pi} \log T, \\ N_0^s(B_{0.3185}) &\ge \left(0.33333816 + o(1)\right) \frac{T}{2\pi} \log T, \end{aligned} \tag{1.5}$$

and

$$\begin{aligned} N_s(B_{0.001}) &\ge \left(0.67250064 + o(1)\right) \frac{T}{2\pi} \log T, \\ N_0(B_{0.001}) &\ge \left(0.67250064 + o(1)\right) \frac{T}{2\pi} \log T, \\ N_0^s(B_{0.001}) &\ge \left(0.34500129 + o(1)\right) \frac{T}{2\pi} \log T. \end{aligned} \tag{1.6}$$

We prove Theorem 2 in the last section of this paper, and there we also include Table 1 which consists of similar results obtained for small values of b. The table also exhibits the fact that the results deteriorate as b increases. (The method ultimately fails when $b \ge 4.2$ for (1.2) and (1.3) and when $b \ge 2$ in (1.4).) It may seem counterintuitive that increasing the width b of the box B_b degrades the results, but the method requires positivity supplied by a kernel that is evaluated on the differences $\rho - \rho'$ of pairs of zeros in B_b. The kernel puts less weight on pairs $\rho = \rho'$ as b increases, which in turn decreases the lower bounds obtained in Theorem 2.

2. Unconditional Montgomery Theorem

In [Mon73], Montgomery introduced the subject of pair correlation of zeros of the zeta-function. He studied, for $x > 0$ and $T \ge 3$,

$$F(x,T) := \sum_{\substack{\rho,\rho' \\ 0<\gamma,\gamma'\le T}} x^{i(\gamma-\gamma')} w(\gamma-\gamma'), \qquad \text{where} \qquad w(u) := \frac{4}{4+u^2}. \tag{2.1}$$

Here the zeros are counted with multiplicity. While the definition of $F(x,T)$ does not depend on RH, the asymptotic formula Montgomery obtained for $F(x,T)$ when $1 \le x \le T$ does depend on RH. This dependence, however, is only because RH allowed the real part of the zeros β to be removed out of $F(x,T)$. In [BGSTB24] we retained this dependence to obtain an unconditional form of Montgomery's theorem. To state our result, let

$$\mathcal{F}(x,T) := \sum_{\substack{\rho,\rho' \\ T<\gamma,\gamma'\leq 2T}} x^{\rho-\rho'} W(\rho-\rho'), \qquad \text{where} \qquad W(u) := \frac{4}{4-u^2}. \tag{2.2}$$

Montgomery Theorem (MT). *For $x \geq 1$ and $T \geq 3$, we have $\mathcal{F}(x,T) \geq 0$, $\mathcal{F}(x,T) = \mathcal{F}(1/x,T)$, and*

$$\mathcal{F}(x,T) = \frac{T}{2\pi x^2}\log^2 T\left(1+O\left(\frac{1}{\sqrt{\log T}}\right)\right) + \frac{T}{2\pi}\log x + O(T\sqrt{\log T}), \tag{2.3}$$

uniformly for $1 \leq x \leq T$.

The statement above has been modified from its original formulation in [BGSTB24], with two changes. First, note that the sum $\mathcal{F}(x,T)$ above is taken over zeros satisfying $T < \gamma,\gamma' \leq 2T$, where as in [BGSTB24] the sums is taken over zeros satisfying $0 < \gamma,\gamma' \leq T$. Secondly, and more importantly, the error terms appearing above have been corrected from those in the original statement of the theorem.* The result stated in [BGSTB24], in contrast to (2.3) above, was

$$\mathcal{F}(x,T) = \left(\frac{T}{2\pi x^2}\log^2 T + \frac{T}{2\pi}\log x\right)\left(1+O\left(\frac{1}{\sqrt{\log T}}\right)\right).$$

Note that if, for example, $x = c\log T$, then

$$\mathcal{F}(x,T) = \frac{T}{2\pi}\log\log T + \frac{T}{2\pi}\left(\frac{1}{c^2}+\log c\right) + O\left(T\frac{\log\log T}{\sqrt{\log T}}\right),$$

and the factor of c on the T term is incorrect. In the corrected version above the error term $O(T\sqrt{\log T})$ which now holds over the whole range $1 \leq x \leq T$ absorbs all these terms in this range. We thank Ramūnas Garunkštis and Julija Paliulionytė for pointing out the mistake and for suggesting an argument to correct the issue. At the beginning of the next section, and before we prove Theorem 1, we explain how to correct the proof of [BGSTB24].

Before we prove **MT**, we record the following standard results. The Riemann von Mangoldt formula for $N(T)$ states that

$$N(T) := \sum_{0<\gamma\leq T} 1 = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T). \tag{2.4}$$

From (2.4), we immediately have

$$N(T) = \frac{T}{2\pi}\log T + O(T) \qquad \text{and} \qquad N(T+1)-N(T) \ll \log T. \tag{2.5}$$

Thus, for example, we see that the sum on the right-hand side of (1.2), (1.3), and (1.4) is

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} 1 = N(2T)-N(T) = \frac{T}{2\pi}\log T + O(T). \tag{2.6}$$

*All the applications of Theorem 1 in [BGSTB24], such as Lemma 5 and Lemma 7, remain correct and analogous to those applications obtained using (2.3). The source of the error came from applying Lemma 8 of [GM87], which has the same issue but likewise no effect on any consequential results in that paper. We note that Montgomery and Vaughan (to appear) have obtained on RH a more refined version of Theorem 1 which deals correctly with the lower order terms which we absorb into a larger error term here.

Using the second estimate in (2.5), or directly as in [Dav00, Lemma, Ch. 15], we have

$$\sum_{\rho} \frac{1}{1+(t-\gamma)^2} \ll \log(|t|+2). \tag{2.7}$$

Using this estimate, we obtain for $w(\gamma-\gamma')$ in (2.1),

$$\sum_{\substack{\rho,\rho' \\ T<\gamma,\gamma'\le 2T}} w(\gamma-\gamma') \ll \sum_{\substack{\rho \\ 0<\gamma\le 2T}} \sum_{\gamma'} \frac{1}{1+(\gamma-\gamma')^2} \ll \sum_{0<\gamma\le 2T} \log\gamma \le N(2T)\log(2T) \ll T\log^2 T. \tag{2.8}$$

3. Proof of MT

In this section we first correct the proof in [BGSTB24, Theorem 1] to obtain Theorem 1 when $1 \le \gamma, \gamma' \le T$. Next, using this result, we prove Theorem 1 for $\mathcal{F}(x,T)$ when $T < \gamma, \gamma' \le 2T$. (This could also be done by modifying each step in the first proof.)

Proof of Theorem 1 in [BGSTB24]. Let I be an interval and

$$\mathcal{F}_I(x) := \sum_{\substack{\rho,\rho' \\ \gamma,\gamma'\in I}} x^{\rho-\rho'} W(\rho-\rho'). \tag{3.1}$$

Thus we see from (2.2) that $\mathcal{F}(x,T) = \mathcal{F}_{(T,2T]}(x)$. Then by (2.17), (2.18) of [BGSTB24] we have, for $1 \le x \le T$,

$$\mathcal{F}_{(0,T]}(x) + O(T^{1/2}) + O(x) = \frac{1}{2\pi} R(x,T), \tag{3.2}$$

where by [BGSTB24, Lemma 1] and following the argument in [Mon73],

$$R(x,T) = \int_0^T |A_1(x,t)+A_2(x,t)+A_3(x,t)|^2 \, dt,$$

where

$$A_1(x,t) = \frac{\log(|t|+2)}{x}, \qquad A_2(x,t) = -\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\frac{1}{2}+it}} \min\left\{\frac{n}{x}, \frac{x}{n}\right\},$$
$$A_3(x,t) \ll \frac{1}{x} + \frac{x^{\frac{1}{2}}}{(1+t^2)} + \frac{x^{-\frac{5}{2}}}{|t|+2}.$$

Now, for $1 \le x \le T$,

$$M_1 := \int_0^T |A_1(x,t)|^2 \, dt = \frac{T}{x^2}(\log^2 T + O(\log T)),$$

and by [GM87, Lemma 7] or [Gol81, Lemma B],

$$M_2 := \int_0^T |A_2(x,t)|^2 \, dt = T\log x + O(T\sqrt{\log T}),$$

and

$$M_3 := \int_0^T |A_3(x,t)|^2 \, dt \ll \frac{T}{x^2} + x.$$

Using Cauchy-Schwarz, and an argument of Ramūnas Garunkštis and Julija Paliulionytė [GP24],

$$\begin{aligned} R(x,T) &= \int_0^T |A_1(x,t)+A_2(x,t)|^2 \, dt + O(\sqrt{M_1M_3}) + O(\sqrt{M_2M_3}) + O(M_3) \\ &= M_1 + M_2 + O\left(\left|\int_0^T A_1(x,t)\overline{A_2(x,t)}\, dt\right|\right) + O(T\sqrt{\log T}) + O\left(\frac{T}{x^2}\log T\right). \end{aligned}$$

Since for $n \geq 2$

$$\int_0^T n^{it}\log(t+2)\, dt \ll \frac{\log T}{\log n},$$

we have

$$\int_0^T A_1(x,t)\overline{A_2(x,t)}\, dt \ll \frac{\log T}{x}\sum_{n=2}^{\infty}\frac{\Lambda(n)}{n^{\frac{1}{2}}\log n}\min\left\{\frac{n}{x},\frac{x}{n}\right\} \ll \frac{\log T}{\sqrt{x}\log 2x} \ll \log T.$$

Thus, by (3.2), we have for $1 \leq x \leq T$,

$$\begin{aligned} \mathcal{F}_{(0,T]}(x) &= \frac{T}{2\pi x^2}(\log^2 T + O(\log T)) + \frac{T}{2\pi}\log x + O(T\sqrt{\log T}) \\ &= \frac{T\log^2 T}{2\pi x^2}\left(1 + O\left(\frac{1}{\sqrt{\log T}}\right)\right) + \frac{T}{2\pi}\log x + O(T\sqrt{\log T}), \end{aligned} \tag{3.3}$$

where we have degraded the first error term insignificantly to simplify applications to sums over differences of zeros. Thus we have obtained **MT** for $\mathcal{F}_{(0,T]}(x)$. □

We now turn to proving **MT** for $\mathcal{F}_{(T,2T]}(x)$.

Lemma 1. *We have, for $x > 0$ and $T \geq 3$,*

$$\mathcal{F}_{(T,2T]}(x,T) = \frac{2}{\pi}\int_{-\infty}^{\infty}\left|\sum_{\substack{\rho \\ T<\gamma\leq 2T}}\frac{x^{\rho-1/2}}{1-(\rho-(1/2+it))^2}\right|^2 dt \tag{3.4}$$

and

$$F_{(T,2T]}(x,T) = \frac{2}{\pi}\int_{-\infty}^{\infty}\left|\sum_{\substack{\rho \\ T<\gamma\leq 2T}}\frac{x^{i\gamma}}{1+(t-\gamma)^2}\right|^2 dt. \tag{3.5}$$

This was proved unconditionally as Lemma 3 in [BGSTB24] with the sum over terms with $0 < \gamma \leq T$. That proof works line by line for both (3.4) and (3.5).

Proof of (2.3). From (3.4) of Lemma 1 we see immediately that $\mathcal{F}_{(T,2T]}(x,T)$ is real and non-negative, and since we can interchange ρ and ρ' in (2.2), we have $\mathcal{F}_{(T,2T]}(1/x,T) = \mathcal{F}_{(T,2T]}(x,T)$.

Thus, $\mathcal{F}(T^\alpha, T)$ is real, non-negative, and even. From (2.2) we have

$$\begin{aligned}\mathcal{F}_{(T,2T]}(x,T) &= \sum_{\substack{\rho,\rho' \\ \gamma,\gamma'\in(T,2T]}} x^{\rho-\rho'} W(\rho-\rho') \\ &= \mathcal{F}_{(0,2T]}(x) - \mathcal{F}_{(0,T]}(x) + O\Bigg(\sum_{\substack{\rho,\rho' \\ 0<\gamma\le T \\ T<\gamma'\le 2T}} x^{|\beta-\beta'|} |W(\rho-\rho')| \Bigg) \\ &= \frac{T\log^2 T}{2\pi x^2}\left(1+O\left(\frac{1}{\sqrt{\log T}}\right)\right) + \frac{T}{2\pi}\log x \\ &\quad + O(T\sqrt{\log T}) + O\Bigg(\sum_{\substack{\rho,\rho' \\ 0<\gamma\le T \\ T<\gamma'\le 2T}} x^{|\beta-\beta'|} |W(\rho-\rho')| \Bigg),\end{aligned}$$

where we have applied (3.3) for both $\mathcal{F}_{(0,2T]}(x)$ and $\mathcal{F}_{(0,T]}(x)$. Note that zeros off the critical line in the upper half plane occur in pairs $\rho = \beta + i\gamma$ and $1-\bar{\rho} = 1-\beta+i\gamma$ with $\beta \neq 1/2$. Thus, letting

$$\Theta(t) := \max\{\beta : \rho = \beta + i\gamma,\ 0 < \gamma \le t\}, \tag{3.6}$$

we have

$$\max_{0<\gamma,\gamma'\le T} |\beta-\beta'| = 2(\max_{0<\gamma\le T}\beta) - 1 = 2\Theta(T) - 1.$$

Hence

$$E_1 := \sum_{\substack{\rho,\rho' \\ 0<\gamma\le T \\ T<\gamma'\le 2T}} x^{|\beta-\beta'|}|W(\rho-\rho')| \ll x^{2\Theta(2T)-1} \sum_{\substack{\rho,\rho' \\ 0<\gamma\le T \\ T<\gamma'\le 2T}} |W(\rho-\rho')|.$$

Using (2.5) and the bound

$$|W(\rho-\rho')| \ll w(\gamma-\gamma'), \tag{3.7}$$

which is (5.5) of [BGSTB24], we see that

$$\begin{aligned}E_1 &\ll x^{2\Theta(2T)-1} \sum_{\substack{\rho \\ 0<\gamma\le T}} \Bigg(\sum_{\substack{\rho' \\ T<\gamma'\le 2T}} w(\gamma-\gamma') \Bigg) \\ &\ll x^{2\Theta(2T)-1} \sum_{\substack{\rho \\ 0<\gamma\le T}} \frac{\log T}{(T-\gamma)+1} \\ &\ll x^{2\Theta(2T)-1}\log^3 T.\end{aligned}$$

Korobov [Kor58b, Kor58a] and Vinogradov [Vin58] (independently) proved that there is a zero-free region for points $\sigma + it$ in the region in the complex plane determined by $\sigma > 1-\eta(t) \ge \Theta(t)$ with

$$\eta(t) = \frac{c}{(\log t)^{2/3}(\log\log t)^{1/3}}, \qquad \text{for } t \ge 3,$$

and some constant $c > 0$. (It is well-known that there are no zeros in the region $\sigma \ge 0$ and $0 \le t \le 3$.) Hence $E_1 \ll x^{1-2\eta(2T)}\log^3 T$. Thus, for $T^{1/2} \le x \le T$,

$$E_1 \ll x^{1-2\eta(2T)}\log^3 T \ll x\exp\left(-c\frac{\log x}{(\log x)^{2/3}(\log\log x)^{1/3}}\right)\log^3 x \ll x,$$

while, for $1 \le x \le T^{1/2}$,

$$E_1 \ll x^{1-2\eta(2T)} \log^3 T \ll T^{1/2}$$

since this error term is increasing in x and thus we may take $x = T^{1/2}$ in the previous bound. Since for $1 \le x \le T$, we have $E_1 \ll T^{1/2} + x \ll T \ll T\sqrt{\log T}$, we conclude

$$\mathcal{F}_{(T,2T]}(x,T) = \frac{T\log^2 T}{2\pi x^2}\left(1 + O\left(\frac{1}{\sqrt{\log T}}\right)\right) + \frac{T}{2\pi}\log x + O(T\sqrt{\log T}).$$

□

4. Evaluation of a Sum over differences of zeros by Montgomery's Theorem

We can use Montgomery's theorem to evaluate many sums over the differences $\rho - \rho'$. If RH is assumed then $\rho - \rho' = i(\gamma - \gamma')$ and we can use the usual real Fourier transform. Thus letting r be in L^1, then $\widehat{r}$, the Fourier transform of r, is well defined and if it is also in L^1 we have

$$r(\alpha) = \int_{-\infty}^{\infty} \widehat{r}(t)e(\alpha t)\,dt, \qquad \widehat{r}(t) := \int_{-\infty}^{\infty} r(\alpha)e(-t\alpha)\,d\alpha, \qquad \text{where} \qquad e(u) := e^{2\pi i u}. \tag{4.1}$$

To prove (1.2) on RH, Montgomery [Mon73] used the Fejér kernel

$$j_F(\alpha) = \max\{1 - |\alpha|, 0\}, \qquad \widehat{j}_F(t) = \left(\frac{\sin \pi t}{\pi t}\right)^2. \tag{4.2}$$

For improved numerical results, we will also use the Montgomery-Taylor kernel [Mon75, CG93]

$$j_M(\alpha) = \frac{1}{1-\cos\sqrt{2}}\left(\frac{1}{2\sqrt{2}}\sin(\sqrt{2}j_F(\alpha)) + \frac{1}{2}j_F(\alpha)\cos(\sqrt{2}\alpha)\right), \tag{4.3}$$

with $j_F(\alpha)$ from (4.2), and

$$\widehat{j}_M(w) = \frac{1}{1-\cos\sqrt{2}}\left(\frac{\sin(\frac{1}{2}(\sqrt{2}-2\pi w))}{\sqrt{2}-2\pi w} + \frac{\sin(\frac{1}{2}(\sqrt{2}+2\pi w))}{\sqrt{2}+2\pi w}\right)^2. \tag{4.4}$$

In the formulas that follow we will take the kernel $j(\alpha)$ to be either $j = j_F$ or $j = j_M$.

Next, without assuming RH we use the complex Fourier transform. Defining

$$\widehat{K_b}(t) := \frac{j(2\pi t)}{\cosh(2\pi b t)}, \tag{4.5}$$

we have for $z \in \mathbb{C}$,

$$K_b(z) = \int_{-\infty}^{\infty} \widehat{K_b}(t)e(zt)\,dt = \int_{-\frac{1}{2\pi}}^{\frac{1}{2\pi}} \frac{j(2\pi t)}{\cosh(2\pi b t)}e^{2\pi i z t}\,dt = \frac{1}{2\pi}\int_{-1}^{1} \frac{j(\alpha)}{\cosh(b\alpha)}e^{iz\alpha}\,d\alpha. \tag{4.6}$$

Note that $K_b(z)$ is an analytic function for all z since the integral defining it is over a finite interval [Tit39, 2.83]. Also, note that if $b = 0$ then $\widehat{K}_0(t) = j(2\pi t)$. We call $K_b(z)$ a Tsang kernel [Tsa93].

We now take $z = -i(\rho - \rho')\log T$ in (4.6) and have

$$K_b(-i(\rho-\rho')\log T) = \frac{1}{2\pi}\int_{-1}^{1} \frac{j(\alpha)}{\cosh(b\alpha)}T^{\alpha(\rho-\rho')}\,d\alpha.$$

Multiplying by $W(\rho - \rho')$ and summing over pairs of zeros ρ, ρ' with $T < \gamma, \gamma' \le 2T$, we obtain by the definition (2.2) that

$$\sum_{\substack{\rho,\rho' \\ T<\gamma,\gamma'\le 2T}} K_b(-i(\rho-\rho')\log T)W(\rho-\rho') = \frac{1}{2\pi}\int_{-1}^{1} \frac{j(\alpha)}{\cosh(b\alpha)}\mathcal{F}(T^\alpha, T)\,d\alpha. \tag{4.7}$$

By Montgomery's Theorem (2.3) we see that $\mathcal{F}(T^\alpha, T)$ is an even function of α, and the right-hand side of (4.7) is equal to

$$\left(\frac{1}{\pi}\int_0^1 \left(\frac{j(\alpha)}{\cosh(b\alpha)}T^{-2\alpha}\log T\left(1+O\left(\frac{1}{\sqrt{\log T}}\right)\right)+\frac{\alpha j(\alpha)}{\cosh(b\alpha)}\right)d\alpha\right)\frac{T}{2\pi}\log T + O(T\sqrt{\log T}).$$

Since

$$\frac{1}{\pi}\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}T^{-2\alpha}\log T\,d\alpha = \frac{1}{\pi}\left(\int_0^{\frac{\log\log T}{\log T}} + \int_{\frac{\log\log T}{\log T}}^1\right)\frac{j(\alpha)}{\cosh(b\alpha)}T^{-2\alpha}\log T\,d\alpha$$

$$= \left(j(0)+O\left(\frac{\log\log T}{\log T}\right)\right)\frac{1}{\pi}\int_0^{\frac{\log\log T}{\log T}} T^{-2\alpha}\log T\,d\alpha + O\left(\frac{1}{\log T}\right) = \frac{1}{2\pi}j(0)+O\left(\frac{\log\log T}{\log T}\right),$$

we conclude that

(4.8)
$$\sum_{\substack{\rho,\rho' \\ T<\gamma,\gamma'\le 2T}} K_b(-i(\rho-\rho')\log T)W(\rho-\rho') = \frac{1}{2\pi}\left(j(0)+2\int_0^1 \frac{\alpha j(\alpha)}{\cosh(b\alpha)}\,d\alpha + O\left(\frac{1}{\sqrt{\log T}}\right)\right)\frac{T}{2\pi}\log T.$$

5. The Tsang Kernel

To obtain the main properties of the Tsang kernel, we need the following Fourier Transform pair.

Lemma 2. *Let t, x be real variables, and take real numbers y and b with $|y|<b$. Then we have the Fourier Transform pair*

(5.1)
$$h_{y,b}(t) = \frac{\cosh(2\pi y t)}{\cosh(2\pi b t)} \qquad \text{and} \qquad \widehat{h}_{y,b}(x) = \frac{1}{b}\left(\frac{\cos\frac{\pi y}{2b}\cosh\frac{\pi x}{2b}}{\cos\frac{\pi y}{b}+\cosh\frac{\pi x}{b}}\right) > 0.$$

Proof of Lemma 2. By [SS03, Example 3, p. 81], a residue calculation gives

$$\int_{-\infty}^{\infty} \frac{e(-xw)}{\cosh \pi x}\,dx = \frac{1}{\cosh \pi w},$$

so that $1/\cosh \pi x$ is its own Fourier transform. Letting $x = 2bt$ and $w = z/2b$, where z is real, then we have $xw = zt$, and taking the real part of both sides of the above equation, we have

(5.2)
$$\int_{-\infty}^{\infty} \frac{\cos(2\pi z t)}{\cosh(2\pi b t)}\,dt = \frac{1}{2b\cosh\frac{\pi z}{2b}}.$$

While this was derived with z real, if we now take $z = x+iy$ with $|y|<b$ we see the integral is absolutely convergent, and therefore each side of this equation is well defined and equals the same analytic function in the horizontal strip $|y|<b$ in the complex plane.

Recalling the formulas

$$\cos z = \cos(x+iy) = \cos x\cosh y - i\sin x\sinh y, \quad \cosh z = \cosh(x+iy) = \cosh x\cos y + i\sinh x\sin y,$$

we have $\mathrm{Re}(\cos z) = \cos x\cosh y$, $\mathrm{Re}(\cosh z) = \cosh x\cos y$, and

$$\mathrm{Re}\left(\frac{1}{\cosh z}\right) = \frac{\mathrm{Re}(\cosh\bar{z})}{|\cosh z|^2} = \frac{\cosh x\cos y}{\cos^2 y+\sinh^2 x} = \frac{\cos y\cosh x}{\frac{1+\cos 2y}{2}+\frac{\cosh 2x-1}{2}} = \frac{2\cos y\cosh x}{\cos 2y+\cosh 2x}.$$

Then, by the above formulas and using that $h_{y,b}(t)$ is even in the second line below, we have

$$\begin{aligned}
\widehat{h}_{y,b}(x) &= \int_{-\infty}^{\infty} e(-xt) h_{y,b}(t)\, dt \\
&= \int_{-\infty}^{\infty} \frac{\cos(2\pi xt)\cosh(2\pi yt)}{\cosh(2\pi bt)}\, dt \\
&= \mathrm{Re} \int_{-\infty}^{\infty} \frac{\cos(2\pi zt)}{\cosh(2\pi bt)}\, dt \\
&= \mathrm{Re} \frac{1}{2b\cosh\frac{\pi z}{2b}} \\
&= \frac{1}{b}\left(\frac{\cos\frac{\pi y}{2b}\cosh\frac{\pi x}{2b}}{\cos\frac{\pi y}{b} + \cosh\frac{\pi x}{b}}\right) > 0.
\end{aligned}$$

□

Remark 3. Tsang [Tsa93] uses $h_y(t)$ which is the case $b = 1$ of the formula above and obtains $\widehat{h}_y(x)$ from [EMOT54, p. 31, (12)], but this is for half of the Fourier transform formula above after a change of variable. Thus Tsang's $\widehat{h}_y(x)$ needs to be doubled, but this has no effect on anything in his paper since only the property $\widehat{h}_y(x) > 0$ is used there.

We next use this lemma to obtain the following lemma, which Tsang [Tsa93, Lemma 1] proved with $b = 1$.

Lemma 3 (K.-M. Tsang)**.** *For fixed $0 < b \ll 1$, the kernel $K_b(z)$ defined by (4.5) and (4.6) is an even entire function for $z = x + iy$, $x, y \in \mathbb{R}$, and we have*

$$\mathrm{Re}\, K_b(x+iy) > 0 \qquad \textit{for all } x \textit{ and } |y| < b. \tag{5.3}$$

Also, for $z \in \mathbb{C}$, we have

$$K_b(z) \ll \frac{e^{|\mathrm{Im}(z)|}}{1+|z|^2}. \tag{5.4}$$

Proof of Lemma 3. By (4.6), we have

$$K_b(z) = \int_{-\frac{1}{2\pi}}^{\frac{1}{2\pi}} \frac{j(2\pi t)}{\cosh(2\pi bt)} e^{2\pi izt}\, dt = 2\int_0^{\frac{1}{2\pi}} \frac{j(2\pi t)}{\cosh(2\pi bt)} \cos(2\pi zt)\, dt. \tag{5.5}$$

Letting $z = x + iy$ and using $\mathrm{Re}(\cos(u+iv)) = \cos u \cosh v$ for real u and v, we have

$$\mathrm{Re}\, K_b(x+iy) = 2\int_0^{\frac{1}{2\pi}} \frac{j(2\pi t)\cosh(2\pi yt)}{\cosh(2\pi bt)} \cos(2\pi xt)\, dt = 2\int_0^{\frac{1}{2\pi}} j(2\pi t) h_{y,b}(t) \cos(2\pi xt)\, dt. \tag{5.6}$$

Hence, by Lemma 2, for $|y| < b$,

$$\mathrm{Re}\, K_b(x+iy) = \big(j(2\pi t)\, h_{y,b}(t)\big)\widehat{\ }(x) = \widehat{j(2\pi t)} * \widehat{h_{y,b}(t)}(x). \tag{5.7}$$

Since $\widehat{j} \geq 0$ by (4.2) and (4.4) and $\widehat{h}_{y,b} > 0$ by (5.1), we see the convolution is positive and (5.3) follows. We now prove (5.4). By (5.5) with the change of variable $\alpha = 2\pi t$, we have

$$K_b(z) = \frac{1}{\pi}\int_0^1 J_b(\alpha)\cos(z\alpha)\, d\alpha, \qquad J_b(\alpha) := \frac{j(\alpha)}{\cosh(b\alpha)}. \tag{5.8}$$

If $|z| \le 1$, then $|K_b(z)| \ll \int_0^1 j(\alpha)\,d\alpha \ll 1$, which proves the estimate in this range. If $|z| > 1$, then by integration by parts

$$\begin{aligned} K_b(z) &= \frac{\sin z\alpha}{\pi z} J_b(\alpha)\Big|_0^1 - \int_0^1 \frac{\sin z\alpha}{\pi z} J_b'(\alpha)\,d\alpha \\ &= \frac{\cos z\alpha}{\pi z^2} J_b'(\alpha)\Big|_0^1 - \int_0^1 \frac{\cos z\alpha}{\pi z^2} J_b''(\alpha)\,d\alpha \ll \frac{e^{|\mathrm{Im}(z)|}}{|z|^2}. \end{aligned}$$

□

Returning to K_b in (4.8), we see from (5.3) of Lemma 3 that

$$\text{(5.9)}\quad \mathrm{Re}\, K_b(-i(\rho-\rho')\log T) = \mathrm{Re}\, K_b((\gamma-\gamma')\log T - i(\beta-\beta')\log T) > 0 \qquad \text{if} \quad |\beta-\beta'| < \frac{b}{\log T}.$$

We want the real part of every term of the sum in (4.8) to be positive (or at least non-negative), and the assumption in Theorem 1 that $|\beta - \frac{1}{2}| < \frac{b}{2\log T}$ for all zeros with $T < \gamma \le 2T$ clearly implies that for any two of these zeros ρ and ρ', we have $|\beta-\beta'| < \frac{b}{\log T}$.

Lemma 4. *Assume that, for all sufficiently large T, all zeros ρ of the Riemann zeta-function with $T < \gamma \le 2T$ are in B_b for some $0 < b \ll 1$. Then*

$$\text{(5.10)}\quad \sum_{\rho,\rho' \in B_b} \mathrm{Re}\, K_b(-i(\rho-\rho')\log T) = \frac{1}{2\pi}\left(j(0) + 2\int_0^1 \frac{\alpha j(\alpha)}{\cosh(b\alpha)}\,d\alpha + O\left(\frac{1}{\sqrt{\log T}}\right)\right)\frac{T}{2\pi}\log T,$$

where every term in the sum above is positive.

Proof of Lemma 4. Since $W(\rho-\rho')$ is complex-valued we need to remove this weight when $|\beta-\beta'| < \frac{b}{\log T}$ in (4.8) before applying (5.9). Since $W(0) = 1$ and $W(z) - 1 = \frac{z^2}{4-z^2}$, we have by (5.4),

$$\text{(5.11)}\quad \begin{aligned} \sum_{\rho,\rho' \in B_b} K_b(-i(\rho-\rho')\log T)(W(\rho-\rho') - 1) &\ll \sum_{\rho,\rho' \in B_b} \frac{T^{|\beta-\beta'|}}{1 + |\rho-\rho'|^2 \log^2 T} \frac{|\rho-\rho'|^2}{|4-(\rho-\rho')^2|} \\ &\le \frac{1}{\log^2 T} \sum_{T<\gamma,\gamma' \le 2T} \frac{T^{b/\log T}}{|4-(\rho-\rho')^2|} \ll \frac{e^b}{\log^2 T} \sum_{T<\gamma,\gamma'\le 2T} w(\gamma-\gamma') \ll e^b T \ll T, \end{aligned}$$

where we used (2.8) and (3.7) in the last line.

To complete the proof, our assumption in Lemma 4 allows us to insert the condition $|\beta-\beta'| < \frac{b}{\log T}$ into the sum in (4.8), which then allows us to remove $W(\rho-\rho')$ from the terms in the sum with an error $\ll T \ll T\sqrt{\log T}$. Taking the real parts proves (5.10). □

6. Proof of Theorem 1

Proof of Theorem 1. In Lemma 4, taking the "diagonal" terms with $\rho = \rho'$ and using (5.8), we obtain the sum

$$\sum_{\substack{\rho \\ T<\gamma\le 2T}} m_\rho\, \mathrm{Re}\, K_b(0) = \sum_{\substack{\rho \\ T<\gamma\le 2T}} m_\rho \left(\frac{1}{\pi}\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\,d\alpha\right),$$

where m_ρ is the multiplicity of the zero ρ.[§] Similarly, the sum of the "symmetric diagonal terms", which are the terms that have $\rho' = 1 - \overline{\rho}$ and $\beta' = 1 - \beta \neq 1/2$, is

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \operatorname{Re} K_b(-i(2\beta-1)\log T) = \sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \left(\frac{1}{2\pi} \int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)} \left(T^{(2\beta-1)\alpha} + T^{-(2\beta-1)\alpha} \right) d\alpha \right).$$

Since all the terms in the sum in Lemma 4 are positive, these diagonal and symmetric diagonal terms provide a lower bound for the sum, and we obtain

(6.1)
$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} m_\rho \left(\frac{1}{\pi} \int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\, d\alpha \right) + \sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \left(\frac{1}{2\pi} \int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)} \left(T^{(2\beta-1)\alpha} + T^{-(2\beta-1)\alpha} \right) d\alpha \right)$$
$$\leq \frac{1}{2\pi} \left(j(0) + 2 \int_0^1 \frac{\alpha j(\alpha)}{\cosh(b\alpha)}\, d\alpha + o(1) \right) \frac{T}{2\pi} \log T.$$

We now take $j(\alpha) = j_F(\alpha) = 1-\alpha$ for $0 \leq \alpha \leq 1$. Under the assumption in Theorem 1 that $b \to 0$ as $T \to \infty$, we have $\cosh(b\alpha) \to 1$ uniformly for $0 \leq \alpha \leq 1$. Similarly, since $e^{-b\alpha} \leq T^{\pm(2\beta-1)\alpha} \leq e^{b\alpha}$, we have $T^{\pm(2\beta-1)\alpha} \to 1$ uniformly for $0 \leq \alpha \leq 1$ when $b \to 0$ as $T \to \infty$. Thus, we have

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} m_\rho \left(\frac{1}{2\pi} + o(1) \right) + \sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \left(\frac{1}{2\pi} + o(1) \right) \leq \frac{1}{2\pi} \left(1 + 2 \int_0^1 \alpha(1-\alpha)\, d\alpha + o(1) \right) \frac{T}{2\pi} \log T,$$

and we conclude that

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} m_\rho + \sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \leq \left(\frac{4}{3} + o(1) \right) \frac{T}{2\pi} \log T. \tag{6.2}$$

To prove (1.2) we use Montgomery's argument. By (6.2),

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} m_\rho \leq \left(\frac{4}{3} + o(1) \right) \frac{T}{2\pi} \log T,$$

and, on recalling (2.6), we have

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \rho \text{ simple}}} 1 \geq \sum_{\substack{\rho \\ T<\gamma\leq 2T}} (2 - m_\rho) \geq \left(2 - \frac{4}{3} + o(1) \right) \frac{T}{2\pi} \log T = \left(\frac{2}{3} + o(1) \right) \frac{T}{2\pi} \log T. \tag{6.3}$$

To prove (1.3), since

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T}} m_\rho \geq \sum_{\substack{\rho \\ T<\gamma\leq 2T}} 1 = (1 + o(1)) \frac{T}{2\pi} \log T,$$

[§]Note that $\rho = \rho'$ is counted with multiplicity in the sum and also weighted by its multiplicity; for example if $m_\rho = 2$ then $\rho = \rho'$ occurs in 4 ways, but is only counted twice in the sum if not weighted by m_ρ.

we have by (6.2)

$$\sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} m_\rho \le \left(\frac43+o(1)\right)\frac{T}{2\pi}\log T - \sum_{\substack{\rho \\ T<\gamma\le 2T}} m_\rho \le \left(\frac13+o(1)\right)\frac{T}{2\pi}\log T. \tag{6.4}$$

Thus

$$\sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta=\frac12}} 1 = \sum_{\substack{\rho \\ T<\gamma\le 2T}} 1 - \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} 1 \ge (1+o(1))\frac{T}{2\pi}\log T - \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} m_\rho \ge \left(\frac23+o(1)\right)\frac{T}{2\pi}\log T.$$

To prove (1.4), we have by (6.3) and (6.4),

$$\begin{aligned}\sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta=\frac12 \text{ and } \rho \text{ simple}}} 1 = \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \rho \text{ simple}}} 1 - \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12 \text{ and } \rho \text{ simple}}} 1 &\ge \left(\frac23+o(1)\right)\frac{T}{2\pi}\log T - \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} m_\rho \\ &\ge \left(\frac13+o(1)\right)\frac{T}{2\pi}\log T.\end{aligned}$$

□

7. Proof of Theorem 2

In Theorem 2 we no longer assume $b\to 0$ as $T\to\infty$. Then, in the second sum in (6.1), we group each term with $\beta>\frac12$ with its symmetric diagonal term that has $\mathrm{Re}(\rho)=1-\beta$, so that this sum is equal to

$$\sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta>\frac12}} m_\rho \left(\frac{2}{\pi}\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\cosh((2\beta-1)\alpha\log T)\,d\alpha\right).$$

Since $\cosh(x)\ge\cosh(0)=1$ for any real number x, we conclude that this sum is

$$\ge \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta>\frac12}} m_\rho \left(\frac{2}{\pi}\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\,d\alpha\right) = \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} m_\rho \left(\frac{1}{\pi}\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\,d\alpha\right).$$

We thus conclude that in place of (6.2) we have

$$\sum_{\substack{\rho \\ T<\gamma\le 2T}} m_\rho + \sum_{\substack{\rho \\ T<\gamma\le 2T \\ \beta\neq\frac12}} m_\rho \le (\mathcal{C}_b(j)+o(1))\frac{T}{2\pi}\log T, \tag{7.1}$$

where

$$\mathcal{C}_b(j) := \frac{j(0)+2\int_0^1 \frac{\alpha j(\alpha)}{\cosh(b\alpha)}\,d\alpha}{2\int_0^1 \frac{j(\alpha)}{\cosh(b\alpha)}\,d\alpha}.$$

Exactly as in the previous section, we have

$$N_s(B_b) \geq (2 - \mathcal{C}_b(j) + o(1))\,\frac{T}{2\pi}\log T, \qquad N_0(B_b) \geq (2 - \mathcal{C}_b(j) + o(1))\,\frac{T}{2\pi}\log T,$$

$$\sum_{\substack{\rho \\ T<\gamma\leq 2T \\ \beta\neq\frac{1}{2}}} m_\rho \leq (\mathcal{C}_b(j) - 1 + o(1))\,\frac{T}{2\pi}\log T, \qquad N_0^s(B_b) \geq (3 - 2\mathcal{C}_b(j) + o(1))\,\frac{T}{2\pi}\log T. \tag{7.2}$$

We now take $j(\alpha) = j_M(\alpha)$ and by Mathematica it is easy to verify Theorem 2 and compute Table 1 and Table 2.

b	$N_s(B_b)$ or $N_0(B_b)$	$N_0^s(B_b)$
0.001	.67250	.34500
0.2	.67019	.34038
0.4	.66333	.32666
0.6	.65208	.30416
0.8	.63670	.27339
1	.61748	.23496
1.2	.59475	.18951
1.4	.56884	.13768
1.6	.54003	.08007
1.8	.50862	.01724
2	.47485	
2.2	.43894	
2.4	.40109	
2.6	.36149	
2.8	.32027	
3	.27760	
3.2	.23357	
3.4	.18832	
3.6	.14194	
3.8	.09451	
4	.04612	
4.187	.00007	

TABLE 1. Lower bounds obtained using j_M for $N_s(B_b)$ or $N_0(B_b)$ and $N_0^s(B_b)$ under the assumption of Theorem 1.

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b	$N_s(B_b)\|_{j_F}$	$N_s(B_b)\|_{j_M}$
0.001	.66666	.67250
0.2	.66422	.67019
0.4	.65697	.66333
0.6	.64509	.65208
0.8	.62886	.63670
1	.60861	.61748
1.2	.58468	.59475
1.4	.55743	.56884
1.6	.52719	.54003
1.8	.49424	.50862
2	.45887	.47485
2.2	.42130	.43894
2.4	.38176	.40109
2.6	.34043	.36149
2.8	.29747	.32027
3	.25304	.27760
3.2	.20727	.23357
3.4	.16026	.18832
3.6	.11214	.14194
3.8	.06298	.09451
4	.01288	.04612
4.0508	.00022	.03368
4.187	0	.00007

TABLE 2. Comparison of using j_F and j_M for $N_s(B_b)$ (or $N_0(B_b)$).

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